

MRI Image Quality Metrics (IQMs)

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Abstract

This document summarizes commonly used MRI image quality metrics (IQMs) and their mathematical definitions for reproducibility and reporting.

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Introduction

This document summarizes commonly used MRI image quality metrics (IQMs) and their mathematical definitions for reproducibility and reporting.

1 Per-Slice IQMs (Foreground / Background pixels)

All metrics below accept foreground pixels $\mathbf{F}$ (from segmentation) and background pixels $\mathbf{B}$. NaN/Inf inputs yield NaN. For BG-dependent metrics (SNR1, SNR2, SNR4, CNR, CJV, FBER), if background is missing or unusable (empty, zero variance, non-finite), we fall back to foreground homogeneity μ_F/σ_F to avoid artificial NaNs.

1.1 Mean (MEAN)

Foreground mean intensity:

$$\text{MEAN} = \mu_F = \frac{1}{N_F} \sum_i F_i$$

[1]

References

- [1] Amir Reza Sadri et al. “MRQy—An open-source tool for quality control of MR imaging data”. In: *Medical physics* 47.12 (2020), pp. 6029–6038.